

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2018-2019)

MEDICAL : NURTURE PHASE-I & II

TEST # 04

DATE : 08-07-2018

Test Type : Minor

Test Pattern : NEET

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **720**.
5. There are **three** parts in the question paper A,B,C consisting of **Physics, Chemistry** and **Biology** having **45 questions** each in Physics & Chemistry and **90 questions** in Biology part of equal weightage. Each question is allotted 4 (four) marks for **correct** response.
6. **One Fourth** mark will be deducted for indicated incorrect response of each question. **No deduction** from the total score will be made if no response is indicated for an item in the Answer Sheet.
7. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
8. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
9. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
10. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
11. **Do not fold or make any stray marks on the Answer Sheet.**

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PART-I [PHYSICS]

- | | |
|--|--|
| <p>1. The displacement time graphs of two particles A and B are straight lines making angles of respectively 30° and 60° with the time axis. If the velocity of A is v_A and that of B is v_B then the value of $\frac{v_A}{v_B}$ is :</p> <p>(1) $\frac{1}{2}$ (2) $\frac{1}{\sqrt{3}}$ (3) $\sqrt{3}$ (4) $\frac{1}{3}$</p> <p>2. A particle covers half of the circle of radius r. Then the displacement and distance of the particle are respectively :</p> <p>(1) $2\pi r, 0$ (2) $2r, \pi r$
 (3) $\frac{\pi r}{2}, 2r$ (4) $\pi r, r$</p> <p>3. The velocity of a particle is $v = v_0 + gt + ft^2$. If its position is $x = 0$ at $t = 0$, then its displacement after unit time ($t = 1$) is</p> <p>(1) $v_0 + 2g + 3f$ (2) $v_0 + \frac{g}{2} + \frac{f}{3}$
 (3) $v_0 + g + f$ (4) $v_0 + \frac{g}{2} + f$</p> <p>4. A train covers the first half of the distance between two stations with a speed of 40 km/h and the other half with 60 km/h. Then its average speed is :</p> <p>(1) 50 km/hr (2) 48 km/h
 (3) 52 km/h (4) 100 km/h</p> | <p>5. The displacement of a particle is zero, the distance covered :</p> <p>(1) must be zero
 (2) may or may not be zero
 (3) cannot be zero
 (4) depends upon the particle</p> <p>6. An athlete completes one round of a circular track of radius R in 20 seconds. What will be his displacement at the end of 2 minute 20 seconds?</p> <p>(1) Zero (2) $2R$
 (3) $2\pi R$ (4) $7\pi R$</p> <p>7. If x denotes displacement in time t and $x = acost$, then acceleration is :-</p> <p>(1) $a \cos t$ (2) $-a \cos t$
 (3) $a \sin t$ (4) $-a \sin t$</p> <p>8. The velocity - time relation of an electron starting from rest is given by $u = kt$, where $k = 2 \text{ m/s}^2$. The distance traversed in 3 sec is :-</p> <p>(1) 9 m (2) 16 m
 (3) 27 m (4) 36 m</p> <p>9. The position x of a particle varies with time (t) as $x = at^2 - bt^3$. The acceleration at time t of the particle will be equal to zero, where t is equal to :-</p> <p>(1) $\frac{2a}{3b}$ (2) $\frac{a}{b}$ (3) $\frac{a}{3b}$ (4) zero</p> |
|--|--|

SPACE FOR ROUGH WORK

10. A particle moves along a straight line such that its displacement at any time t is given by $s = t^3 - 6t^2 + 3t + 4$ metres. The velocity when the acceleration is zero is :-
 (1) 3 m/s (2) -12 m/s
 (3) 42 m/s (4) -9 m/s
11. The displacement of a particle starting from rest (at $t = 0$) is given by $s = 6t^2 - t^3$. The time when the particle will attain zero velocity again, is :-
 (1) 4 s (2) 8 s
 (3) 12 s (4) 16 s
12. The velocity of a body depends on time according to equation $v = 20 + 0.1 t^2$. The body has :-
 (1) uniform acceleration
 (2) uniform retardation
 (3) non-uniform acceleration
 (4) zero acceleration
13. The displacement of a particle is given by $y = a + bt + ct^2 - dt^4$. The initial velocity and acceleration are respectively :-
 (1) $b, -4d$ (2) $-b, 2c$
 (3) $b, 2c$ (4) $2c, -4d$
14. The initial velocity of a particle is u (at $t = 0$) and the acceleration is given by $f = at$. Which of the following relations is valid?
 (1) $v = u + at^2$ (2) $v = u + \frac{at^2}{2}$
 (3) $v = u + at$ (4) $v = u$
15. The relation $t = \sqrt{x} + 3$ describes the position of a particle where x is in metres and t is in seconds. The position, when velocity is zero, is :-
 (1) 2 m (2) 4 m (3) 5 m (4) zero
16. If for a particle position $x \propto t^2$ then :-
 (1) velocity is constant
 (2) acceleration is constant
 (3) acceleration is variable
 (4) None of these
17. A body is moving according to the equation $x = at + bt^2 - ct^3$. Then its instantaneous speed is given by :-
 (1) $a + 2b + 3ct$ (2) $a + 2bt - 3ct^2$
 (3) $2b - 6ct$ (4) none of these
18. A body covered a distance of 5 m along a semicircular path. The ratio of distance to displacement is :-
 (1) 11 : 7 (2) 12 : 5 (3) 8 : 3 (4) 7 : 5
19. A body goes 10 km north and 20 km east. What will be the displacement from initial point?
 (1) 22.36 km (2) 2 km
 (3) 5 km (4) 20 km
20. A scooter is going towards east at 10 m s^{-1} turns right through an angle of 90° . If the speed of the scooter remains unchanged in taking this turn, the change in the velocity of the scooter is :-
 (1) 20.0 ms^{-1} in south-western direction
 (2) zero
 (3) 10.0 ms^{-1} in southern direction
 (4) 14.14 ms^{-1} in south-western direction

SPACE FOR ROUGH WORK

21. A particle experiences constant acceleration for 20 seconds after starting from rest. If it travels a distance s_1 in the first 10 seconds and distance s_2 in the next 10 seconds, then :-
 (1) $s_2 = s_1$ (2) $s_2 = 2s_1$
 (3) $s_2 = 3s_1$ (4) $s_2 = 4s_1$
22. Two balls of different masses m_a and m_b are dropped from two different heights, viz, a and b. The ratio of times taken by the balls to drop through these distances is :-
 (1) $a : b$ (2) $b : a$
 (3) $\sqrt{a} : \sqrt{b}$ (4) $a^2 : b^2$
23. If two balls of same density but different masses are dropped from a height of 100 m, then :-
 (1) both will come together on the earth
 (2) both will come late on the earth
 (3) first will come first and second after that
 (4) second will come first and first after that
24. The splash is heard after 2.05 s after the stone is dropped into a well of depth 19.6 m. The velocity of sound is :-
 (1) 342 ms^{-1} (2) 372 ms^{-1}
 (3) 392 ms^{-1} (4) 352 ms^{-1}
25. An engine of a train moving with uniform acceleration passes an electric pole with velocity u and the last compartment with velocity v . The middle point of the train passes past the same pole with a velocity of :-
 (1) $\frac{u+v}{2}$ (2) $\frac{u^2+v^2}{2}$
 (3) $\sqrt{\frac{u^2+v^2}{2}}$ (4) $\sqrt{\frac{u^2-v^2}{2}}$
26. The acceleration of a particle, starting from rest, varies with time according to the relation :
 $a = -\omega^2 \sin \omega t$
 The displacement of this particle at a time t will be :-
 (1) $s \sin \omega t$ (2) $s \omega \cos \omega t$
 (3) $s \omega \sin \omega t$ (4) $\frac{1}{2}(s \omega^2 \sin \omega t)t^2$
27. A ball is released from the top of a tower of height h metres. It takes T seconds to reach the ground. What is the position of the ball in $T/3$ seconds?
 (1) $h/9$ metres from the ground
 (2) $7h/9$ metres from the ground
 (3) $8h/9$ metres from the ground
 (4) $17h/18$ metres from the ground
28. A particle is moving eastward with a velocity of 5 m/second. In 10 seconds, the velocity changes to 5 m/second northward. The average acceleration in this time is :-
 (1) $\frac{1}{\sqrt{2}} \text{ m/sec}^2$ towards north-west
 (2) $\frac{1}{\sqrt{2}} \text{ m/sec}^2$ towards north-east
 (3) $\frac{1}{2} \text{ m/sec}^2$ towards north-west
 (4) $\frac{1}{2} \text{ m/sec}^2$ towards north-east

SPACE FOR ROUGH WORK

29. The relation between time t and distance x is

$$t = \alpha x^2 + \beta x$$

where α and β are constants. The retardation is :-

- (1) $2\alpha v^3$ (2) $2\beta v^3$
 (3) $2\alpha\beta v^3$ (4) $2\beta^2 v^3$
30. When the speed of a car is v , the minimum distance over which it can be stopped is s . If the speed becomes nv , what will be the minimum distance over which it can be stopped during same time?
- (1) s/n (2) ns
 (3) s/n^2 (4) n^2s
31. Dimension of gravitational constant are :-
- (1) $[ML^2T^2]$ (2) $[M^1L^3T^{-2}]$
 (3) $[M^0L^3T^2]$ (4) $[M^{-1}L^3T^{-2}]$
32. The dimensional formula for kinetic energy is :-
- (1) $[M^2L^2T]$ (2) $[ML^2T^{-2}]$
 (3) $[M^0L^{-1}]$ (4) $[ML^2T]$
33. The velocity v (in cm/sec) of a particle is given in terms of time t (in sec) by the equation

$$v = at + \frac{b}{t+c}$$

The dimensions of a , b and c are :-

- | a | b | c |
|-----------------|--------|----------|
| (1) $[L^2]$ | $[T]$ | $[LT^2]$ |
| (2) $[LT^2]$ | $[LT]$ | $[L]$ |
| (3) $[LT^{-2}]$ | $[L]$ | $[T]$ |
| (4) $[L]$ | $[LT]$ | $[T^2]$ |

34. In the relation :-

$$y = a \sin (\omega t - Kx)$$

the dimensional formula for K is :

- (1) $[M^0LT]$ (2) $[M^0L^{-1}T^0]$
 (3) $[M^0LT^{-1}]$ (4) $[M^0L^{-1}T^{-1}]$
35. In the relation : $y = a \cos (\omega t + Kx)$, the dimensional formula for Kx is same as that of:
- (1) $\frac{a}{\omega}$ (2) $\frac{a}{y}$ (3) $\frac{\omega t}{a}$ (4) $\frac{ya}{\omega t}$
36. If the speed of light (c), acceleration due to gravity (g) and pressure (p) are taken as fundamental units, the dimensions of gravitational constant (G) are :
- (1) $c^0 g p^{-3}$
 (2) $c^2 g^3 p^{-2}$
 (3) $c^0 g^2 p^{-1}$
 (4) $c^2 g^2 p^{-2}$
37. If the velocity of light (c), gravitational constant (G) and Planck's constant (h) are chosen as fundamental units, then which of the following represents the dimensions of the mass ?
- (1) $[c^{1/2} G^{1/2} h^{1/2}]$ (2) $[c^{1/2} G^{-1/2} h^{-1/2}]$
 (3) $[c^{1/2} G^{-1/2} h^{1/2}]$ (4) $[c^{-1/2} G^{1/2} h^{-1/2}]$
38. If energy (E), velocity (v) and force (F) be taken as fundamental quantities, then what are the dimensions of mass ?
- (1) $[Ev^2]$ (2) $[Ev^{-2}]$
 (3) $[Fv^{-1}]$ (4) $[Fv^{-2}]$

SPACE FOR ROUGH WORK

39. A student measured the diameter of a wire using a screw gauge with least count 0.001 cm and listed the measurements. The correct measurement is :
- (1) 5.320 cm
 - (2) 5.3 cm
 - (3) 5.32 cm
 - (4) 5.3200 cm
40. An experiment measured quantities a, b, c and then x calculated from $x = \frac{ab^2}{c^3}$. If the percentage errors a, b, c are $\pm 1\%$, $\pm 3\%$ and $\pm 2\%$ respectively, the percentage error in x can be:
- (1) $\pm 13\%$
 - (2) $\pm 7\%$
 - (3) $\pm 4\%$
 - (4) $\pm 1\%$
41. The resultant of $\vec{A} + \vec{B}$ is $\vec{R}_1$. On reversing the vector $\vec{B}$, the resultant $\vec{R}_2$. What is the value of $R_1^2 + R_2^2$?
- (1) $A^2 + B^2$
 - (2) $A^2 - B^2$
 - (3) $2(A^2 + B^2)$
 - (4) $2(A^2 - B^2)$
42. A particle moves through angular displacement θ on a circular path of radius r. The linear displacement will be :
- (1) $2r \sin (\theta/2)$
 - (2) $2r \cos (\theta/2)$
 - (3) $2r \tan (\theta/2)$
 - (4) $2r \cot (\theta/2)$
43. A vector of length l is turned through the angle θ about its tail. What is the change in the position vector of its head ?
- (1) $l \cos (\theta/2)$
 - (2) $2l \sin (\theta/2)$
 - (3) $2l \cos (\theta/2)$
 - (4) $l \sin (\theta/2)$
44. Given that $A = B$. What is the angle between $\vec{A} + \vec{B}$ and $\vec{A} - \vec{B}$?
- (1) 30°
 - (2) 60°
 - (3) 90°
 - (4) 180°
45. Given that $\vec{A} + \vec{B} = \vec{R}$ and $\vec{A} + 2\vec{B}$ is perpendicular to $\vec{A}$. Then :
- (1) $2B = R$
 - (2) $B = 2R$
 - (3) $B = R$
 - (4) $B^2 = 2R^2$

SPACE FOR ROUGH WORK

PART-II [CHEMISTRY]

46. The triad of nuclei that is isotonic :-
 (1) ${}^{14}_6\text{C}$, ${}^{15}_7\text{N}$, ${}^{17}_9\text{F}$ (2) ${}^{14}_6\text{C}$, ${}^{14}_7\text{N}$, ${}^{17}_9\text{F}$
 (3) ${}^{13}_6\text{C}$, ${}^{14}_7\text{N}$, ${}^{19}_9\text{F}$ (4) ${}^{14}_6\text{C}$, ${}^{14}_7\text{N}$, ${}^{19}_9\text{F}$
47. The wavelength associated with a golf ball weighing 200 g and moving at a speed of 5 m/h is of the order of :-
 (1) 10^{-1} m (2) 10^{-20} m
 (3) 10^{-30} m (4) 10^{-40} m
48. For principle quantum number $n = 4$, the total number of orbitals having $\ell = 3$ is :-
 (1) 3 (2) 7
 (3) 5 (4) 9
49. Choose the wrong one on the basis of Bohr's theory:-
 (1) Velocity of electron $\propto \frac{1}{\sqrt{r}}$
 (2) Frequency of revolution $\propto \frac{1}{n^3}$
 (3) Velocity of electron $\propto \sqrt{r}$
 (4) $mvr \propto n$
50. If kinetic energy of a proton is increased nine times the wavelength of the de-Broglie wave associated with it would become :-
 (1) 3 times (2) 9 times
 (3) $\frac{1}{3}$ times (4) $\frac{1}{9}$ times
51. Among the following groups which represents the collection of isoelectronic species ?
 (1) NO^+ , C_2^{2-} , O_2^- , CO (2) N_2 , C_2^{2-} , CO , NO
 (3) CO , NO^+ , CN^- , C_2^{2-} (4) NO , CN^- , N_2 , O_2^-
52. Which one of the following sets of quantum numbers represents an impossible arrangement?
- | n | ℓ | m_ℓ | m_s |
|-------|--------|----------|-------|
| (1) 3 | 2 | -2 | + 1/2 |
| (2) 4 | 0 | 0 | + 1/2 |
| (3) 3 | 2 | -3 | + 1/2 |
| (4) 5 | 3 | 0 | - 1/2 |
53. An electron is present in 4f sub-shell. The possible values of azimuthal quantum number for this electron are :-
 (1) 0, 1, 2, 3 (2) 1, 2, 3, 4
 (3) 3 (4) 4
54. How many electrons in calcium have $\ell = 0$?
 (1) 6 (2) 8
 (3) 10 (4) 12
55. For $n = 4$
 (1) The total possible values of ℓ are 3
 (2) The highest value of ℓ is 4
 (3) The total number of possible values of m is 7
 (4) The highest value of m is +3

SPACE FOR ROUGH WORK

56. The maximum number of electrons in p-orbital with $n = 6$, $m = 0$ is :-
(1) 2 (2) 6 (3) 10 (4) 14
57. Number of mols in 1200 gm NH_2CONH_2 are
(1) 10 (2) 20 (3) 30 (4) 40
58. Mass of CO_2 produced by combustion of 24 gm carbon in excess supply of oxygen is
(1) 44 gm (2) 22 gm (3) 88 gm (4) 48 gm
59. Molarity of a solution prepared by dissolving 20 gm NaOH in 1500 mL solution is :
(1) 0.5 M (2) 1.0 M
(3) 0.1 M (4) None of these
60. If molecular mass of $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$ is M, then its equivalent mass will be
(1) M (2) $\frac{M}{2}$ (3) $\frac{M}{3}$ (4) $\frac{M}{4}$
61. Number of gram equivalents in 64 gm O_2 are
(1) 2 (2) 4 (3) 16 (4) 8
62. **Column-I** **Column-II**
A. law of triads 1. Newland
B. Law of octaves 2. Mendeleev
C. Periodic law 3. Moseley
D. Modified periodic law 4. Dobereiner
The correct match of contents in column I with that of column II is :

	A	B	C	D
(1)	1	2	3	4
(2)	4	3	2	1
(3)	1	3	4	2
(4)	4	1	2	3
63. Eka-aluminium and eka-silicon are known as:
(1) aluminium and silicon
(2) gallium and germanium
(3) scandium and titanium
(4) stable isotopes of aluminium and silicon
64. The elements in which electrons are progressively filled in 4f-orbitals are called:
(1) lanthanoids
(2) actinoids
(3) noble gases
(4) chalcogens
65. Which of the elements whose atomic numbers are given below, cannot be accommodated in the present set up of the long form of periodic table?
(1) 107 (2) 118
(3) 126 (4) 102
66. An element having electronic configuration $[\text{Ar}]3d^2, 4s^2$ belongs to :
(1) d-block (2) f-block
(3) s-block (4) p-block
67. The electronic configuration of most electropositive element is :
(1) $[\text{Xe}] 6s^2$ (2) $[\text{Xe}] 6s^1$
(3) $[\text{Kr}] 5s^1$ (4) $[\text{Kr}] 5s^2$
68. An element which belongs to third period and group 15 has electronic configuration :
(1) $1s^2, 2s^2 2p^6, 3s^2 3p^3$
(2) $1s^2, 2s^2 2p^3$
(3) $1s^2, 2s^2 2p^6, 3s^2 3p^4$
(4) None of these

SPACE FOR ROUGH WORK

69. d-block elements in long form of periodic table are placed:
(1) on the extreme left
(2) on the extreme right
(3) at the bottom
(4) in the middle
70. The element of 4th period of the periodic table having maximum number of unpaired electrons in its ground state is :
(1) ${}_{20}\text{Ca}$ (2) ${}_{30}\text{Zn}$ (3) ${}_{24}\text{Cr}$ (4) ${}_{28}\text{Ni}$
71. The screening effect of d-electrons is :
(1) much less than s-electrons
(2) much more than s-electrons
(3) equal to s-electrons
(4) equal to p-electrons
72. The first ionisation enthalpies of Na, Mg, Al and Si are in the order of :
(1) $\text{Na} < \text{Mg} > \text{Al} < \text{Si}$
(2) $\text{Na} > \text{Mg} > \text{Al} > \text{Si}$
(3) $\text{Na} < \text{Mg} < \text{Al} > \text{Si}$
(4) $\text{Na} > \text{Mg} > \text{Al} < \text{Si}$
73. Match List-I with List-II and select the correct answer :
- | List-I(At.No) | List-II(Block) |
|---------------|----------------|
| A. 43 | 1. s-block |
| B. 35 | 2. p-block |
| C. 38 | 3. d-block |
| D. 60 | 4. f-block |
- (1) A-2, B-1, C-3, D-4
(2) A-1, B-2, C-3, D-4
(3) A-4, B-3, C-2, D-1
(4) A-3, B-2, C-1, D-4
74. Which one of the following grouping represent a collection of isoelectronic species ?
(At. numbers, Cs = 55 and Br = 35)
(1) N^{3-} , F^- , Na^+
(2) Ca^{2+} , Cs^+ , Br
(3) Be, Al^{3+} , Cl^-
(4) Na^+ , Ca^{2+} , Mg^{2+}
75. Which of the following elements has the maximum electron gain enthalpy ?
(1) Oxygen (2) Chlorine
(3) Fluorine (4) Nitrogen
76. The set representing the correct order of first ionisation enthalpy is :
(1) $\text{K} > \text{Na} > \text{Li}$
(2) $\text{Be} > \text{Mg} > \text{Ca}$
(3) $\text{B} > \text{C} > \text{N}$
(4) $\text{Ge} > \text{Si} > \text{C}$
77. The correct order of atomic radii or ionic radii is:-
(1) $\text{Na} < \text{Be} < \text{B}$
(2) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
(3) $\text{Na} < \text{Li} < \text{K}$
(4) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
78. Which of the following is the smallest cation?
(1) Na^+ (2) Mg^{2+}
(3) Al^{3+} (4) Ca^{2+}
79. The ionic radius of a cation is always
(1) less than atomic radius
(2) more than atomic radius
(3) equal to atomic radius
(4) cannot be predicted

SPACE FOR ROUGH WORK

- 80.** The elements with atomic number 10, 18, 36, 54 and 86 are all :
(1) light metals
(2) inert gases
(3) halogens
(4) none of these
- 81.** Atoms of which of the following groups lose electrons most easily ?
(1) Li, Na, K
(2) Cl, Br, I
(3) O, S, Se
(4) N, P, As
- 82.** Of the following elements, which one has the highest electronegativity
(1) I (2) Br
(3) Cl (4) F
- 83.** The element with positive value of electron gain enthalpy is :
(1) hydrogen
(2) oxygen
(3) lithium
(4) neon
- 84.** Which of the following has the lowest ionisation enthalpy ?
(1) $4s^1$ (2) $3d^2$
(3) $3p^6$ (4) $2p^6$
- 85.** Aluminium is diagonally related to :
(1) Li (2) Si
(3) Be (4) B
- 86.** Which one is the correct order of the size of the iodine species ?
(1) $I > I^+ > I^-$ (2) $I > I^- > I^+$
(3) $I^+ > I^- > I$ (4) $I^- > I > I^+$
- 87.** In the long form of the periodic table, all the non-metals are placed with :
(1) s-block
(2) p-block
(3) d-block
(4) f-block
- 88.** Which of the following is not correct ?
(1) $N^{3-} > O^{2-} > F^-$: Ionic radii
(2) $s > p > d$: Ionisation energy
(3) $Cl > F > Br$: Electron gain enthalpy
(4) $P > N > O$: Electronegativity
- 89.** The correct sequence which shows decreasing order of ionic radii of the elements is :
(1) $O^{2-} > F^- > Na^+ > Mg^{2+} > Al^{3+}$
(2) $Al^{3+} > Mg^{2+} > Na^+ > F^- > O^{2-}$
(3) $Na^+ > Mg^{2+} > Al^{3+} > O^{2-} > F^-$
(4) $Na^+ > F^- > Mg^{2+} > O^{2-} > Al^{3+}$
- 90.** For one of the element various successive ionization enthalpies (in kJ mol^{-1}) are given below :

IE	1st	2nd	3rd	4th	5th
	577.5	1810	2750	11580	14820

The element is
(1) P (2) Mg
(3) Si (4) Al

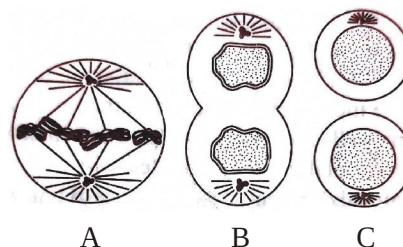
SPACE FOR ROUGH WORK

PART-III [BIOLOGY]

91. What is the approximate percentage duration of cell cycle that comes under interphase in human :
- (1) 100% (2) 95% (3) 25% (4) 5%
92. A somatic cell that has just completed the S-phase of its cell cycle, as compared to gamete of the same species has :
- (1) twice the number of chromosomes and twice the amount of DNA
(2) same number of chromosomes, but twice the amount of DNA
(3) twice the number of chromosomes and four times the amount of DNA
(4) four times the number of chromosomes and twice the amount of DNA
93. Which of the following statements is true for cells in G_0 -stage of cell cycle ?
- (1) Cells in G_0 -stage are metabolically more active
(2) Cells are metabolically inactive
(3) Cells are metabolically active, but no longer proliferate in normal condition
(4) None of the above
94. I. Cells like heart cells and nerve cells, which are also called non-dividing cells are inA.... phase.
II. During S-phase in animal cells, DNA replication takes place in theB..... and duplication of centriole takes place in theC..... .

Fill in the blanks with appropriate option.

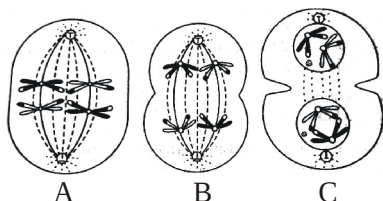
- (1) A- G_0 , B-nucleus, C-cytoplasm
(2) A-S, B-cytoplasm, C-nucleus
(3) A- G_1 , B-nucleus, C-cytoplasm
(4) A- G_0 , B-cytoplasm, C-nucleus
95. This is the best stage to count the number and study the morphology of chromosomes.
- (1) Prophase (2) Metaphase
(3) Anaphase (4) Telophase
96. See the diagrams carefully and identify the different stages of mitosis (A-C) by choosing appropriate options given below.



- (1) A-Metaphase; B-Telophase; C-Interphase
(2) A-Telophase; B-Metaphase; C-Prophase
(3) A-Anaphase; B-Telophase; C-Interphase
(4) A-Telophase; B-Anaphase; C-Prophase
97. Zygotene of prophase-I is characterised by :
- (1) chromosomes
(2) synaptonemal complex
(3) crossing over
(4) terminalisation of chiasmata

SPACE FOR ROUGH WORK

98. Identify the given figures showing meiotic phases and select the correct option.



A B C

- (1) Metaphase Anaphase Telophase
(2) Metaphase-I Anaphase-I Telophase-I
(3) Metaphase-II Anaphase-II Telophase-II
(4) Anaphase-I Metaphase-I Telophase-I

99. Diakinesis is marked by

- (1) terminalisation of chiasmata
(2) degeneration of nucleolus
(3) fully condensed chromosomes
(4) All of the above

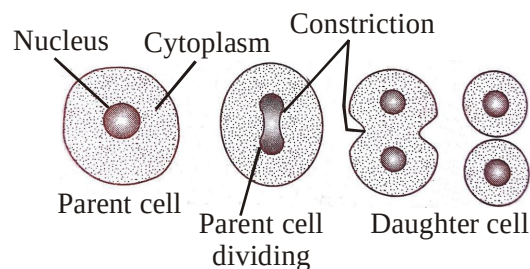
100. Select the correct statements regarding S-phase of interphase.

- i. Occurs between G_1 and G_2
ii. DNA replication begins in the nucleus
iii. Centrioles duplicate in the cytoplasm
iv. As DNA is doubled, number of chromosomes also doubles

The option with correct statements is :

- (1) iv and iii
(2) i, ii, iii and iv
(3) ii, iii and iv
(4) i, ii and iii

101. See the diagram carefully and sequentially arrange the steps of amitosis given below.



- i. The constriction appears in the cytoplasm.
ii. The nucleus of cell elongates and develops a constriction around its middle.
iii. The constriction in nucleus gradually deepens and finally cuts the nucleus into two daughter nuclei.
iv. Then cytoplasmic constriction divides the parent cell into two daughter cell, each with a nucleus.

Option containing correct sequence of events is

- (1) i $\rightarrow$ iii $\rightarrow$ ii $\rightarrow$ iv
(2) i $\rightarrow$ ii $\rightarrow$ iii $\rightarrow$ iv
(3) ii $\rightarrow$ i $\rightarrow$ iii $\rightarrow$ iv
(4) ii $\rightarrow$ iii $\rightarrow$ i $\rightarrow$ iv

SPACE FOR ROUGH WORK

102. While in mitosis, the daughter cells resemble each other and also the parent cell; in meiosis they differ not only from parent cell in having half the number of chromosomes, but also differ among themselves qualitatively in genetic constitution due to :

- (1) Segregation and crossing over only
- (2) independent assortment and segregation only
- (3) independent assortment and crossing over only
- (4) crossing over, independent assortment and segregation

103. Match the following columns.

Column I	Column II
A. Terminalisation	1. Metaphase-I
B. Histone synthesis	2. Anaphase-I
C. Disjunction	3. Diakinesis
D. Interkinesis	4. Meiosis-II
E. Double equatorial plate	5. S-phase

Codes

	A	B	C	D	E
(1)	5	2	1	3	4
(2)	3	5	4	2	1
(3)	2	5	4	1	3
(4)	3	5	2	4	1

104. Find examples where the four daughter cells from meiosis are equal in size (I) and where they are found unequal in size (II).

- (1) I. Microsporogenesis
II. Megasporogenesis
- (2) I. Spermatogenesis
II. Oogenesis
- (3) Both (1) and (2)
- (4) None of the above

105. A bivalent meiosis-I consists of

- (1) two chromatids and one centromere
- (2) two chromatids and two centromeres
- (3) four chromatids and two centromeres
- (4) four chromatids and four centromeres

106. Systematic botany refers to the

- (1) system analysis
- (2) systematic arrangement of organs of plants
- (3) systematic study of organelles and tissues
- (4) methodical study of plants, dealing with identification, nomenclature and classification

107. Arrange the following in ascending order of similar characteristics.

- I. Family II. Genus
- III. Class IV. Species
- (1) Class < Family < Genus < Species
- (2) Family < Class < Genus < Species
- (3) Genus < Order < Species < Class
- (4) Class < Genus < Species < Family

SPACE FOR ROUGH WORK

- | | |
|---|--|
| <p>108. The scientific name of wheat is :</p> <ol style="list-style-type: none"> (1) <i>Mangifera indica</i> (2) <i>Triticum aestivum</i> (3) <i>Triticum poales</i> (4) None of these <p>109. Largest herbarium in India is :</p> <ol style="list-style-type: none"> (1) Madras Herbarium, Coimbatore (Tamil Nadu) (2) Central National Herbarium (Indian Botanical Garden) Sibpur, Kolkata (WB) (3) Herbarium of National Botanical Research Institute, Lucknow (UP) (4) Forest Research Institute, Dehradun (UK) <p>110. which one of the following is not a correct statement ?</p> <ol style="list-style-type: none"> (1) Herbarium houses dried, pressed and preserved plant specimens (2) Botanical gardens have collection of living plants for reference (3) A museum has collection of photographs of plants and animals (4) Key is a taxonomic aid for identification of specimens <p>111. Which taxonomical aid provides all information about a particular taxon like order or family ?</p> <ol style="list-style-type: none"> (1) Herbarium (2) Catalogue (3) Taxonomic key (4) Monograph | <p>112. As we go from species to kingdom in a taxonomic hierarchy, the number of common characteristics :</p> <ol style="list-style-type: none"> (1) will decrease (2) will increase (3) remain same (4) may increase or decrease <p>113. Which of the following characters served as the criteria for five kingdom system of classification proposed by Whittaker ?</p> <ol style="list-style-type: none"> (1) Cell structure (2) Body organisation and mode of nutrition (3) Reproduction and phylogenetic relationships (4) All of the above <p>114. In the light of recent classification of living organisms into three domains of life (bacteria, archaea and eukarya), which one of the following statements is true about archaea ?</p> <ol style="list-style-type: none"> (1) Archaea resemble eukarya in all respects (2) Archaea have some noble features that are absent in other prokaryotes and eukaryotes (3) Archaea completely differ from both prokaryotes and eukaryotes (4) Archaea completely differ from prokaryotes <p>115. Which of the following bacteria play an important role in the recycling of nutrients like nitrogen, phosphorus, iron and sulphur ?</p> <ol style="list-style-type: none"> (1) Chemoheterotrophic bacteria (2) Chemosynthetic autotrophic bacteria (3) Parasitic bacteria (4) Saprophytic bacteria |
|---|--|

SPACE FOR ROUGH WORK

- 116.** Citrus canker is a
 (1) viral disease
 (2) bacterial disease
 (3) fungal disease
 (4) protozoan disease
- 117.** Select wrong statement.
 (1) The viroids were discovered by DJ Ivanowski
 (2) WM Stanley showed that viruses could be crystallised
 (3) The term '*contagium vivum fluidum*' was coined by MW Beijerinck
 (4) Mosaic disease in tobacco and AIDS in human being are caused by viruses
- 118.** Viroids have :
 (1) ds DNA enclosed by protein coat
 (2) ss DNA enclosed by protein coat
 (3) ss RNA not enclosed by protein coat
 (4) ds RNA enclosed by protein coat
- 119.** The latest view for the origin of viruses is :
 (1) they have arisen from nucleic acid and proteins found in primitive soup
 (2) they arose from bacteria as a result of the loss of cell wall, ribosome, etc.
 (3) they arose from some bacteria, which had developed a nucleus only
 (4) they are modified plasmids, which are infact the fragments of the nucleic acids of the host
- 120.** TO Diener (1971) discovered a new infectious agent that was smaller than viruses. Consider the following statements about this infectious agent.
 I. It causes potato spindle tuber disease.
 II. These are infectious RNA particles.
 III. It lacks a protein coat.
 IV. The molecular weight of its RNA is low.
 The above statements are assigned to
 (1) viruses
 (2) viroids
 (3) prions
 (4) lichen
- 121.** Match the following columns.
- | Column-I | Column-II |
|-------------------------|-----------------------------------|
| A. DJ Ivanowsky (1892) | 1. Viroids |
| B. MW Beijerinck (1898) | 2. First crystallised TMV |
| C. WM Stanley (1935) | 3. <i>contagium vivum fluidum</i> |
| D. TO Diener (1971) | 4. Mosaic disease of tobacco |
- Codes :**
- | | A | B | C | D |
|-----|---|---|---|---|
| (1) | 1 | 4 | 3 | 2 |
| (2) | 2 | 1 | 4 | 3 |
| (3) | 4 | 3 | 2 | 1 |
| (4) | 3 | 2 | 1 | 4 |

SPACE FOR ROUGH WORK

- 122.** Which of the following set of organisms reproduces by fragmentation (asexual mode of reproduction) ?
- (1) *Amoeba*, fungi and earthworm
 - (2) Fungi, filamentous algae and protonema of mosses
 - (3) *Hydra*, fungi, *Amoeba* and bacteria
 - (4) Earthworm, bacteria and fungi
- 123.** Identify the incorrect statement given below.
- (1) Cellular organisation of the body is the defining feature of non-living forms
 - (2) Consciousness is the property shared by all living organisms
 - (3) A patient with dead brain has no self consciousness
 - (4) Human beings are the only organisms, who is aware of himself, i.e. self conscious
- 124.** Choose the incorrect statement regarding universal rules of nomenclature.
- (1) The first word of a biological name represents the genus
 - (2) The first word of the genus starts with a capital letter
 - (3) Both the words in a biological name, when handwritten are separately underlined
 - (4) Biological names are generally in English and are printed in italics
- 125.** Which of the following is not the basis of five kingdom system of classification proposed by R.H. Whittaker ?
- A. Phylogenetic relationship
 - B. Cell structure
 - C. Thallus organisation
 - D. Reserve food material
 - E. Photosynthetic pigments
 - F. Reproduction
- (1) D only
 - (2) A, D and F
 - (3) A, D and E
 - (4) D and E
- 126.** Three domains of life are given on the basis of
- (1) Cell wall characters
 - (2) Presence or absence of 16-S rRNA genes
 - (3) 16-S rRNA base sequencing
 - (4) Mode of nutrition
- 127.** The process of genetic recombination in which DNA is transferred from a bacterium to another bacterium by physical contact is known as
- (1) Transformation
 - (2) Conjugation
 - (3) Specialised transduction
 - (4) Generalised transduction
- 128.** Sexduction occurs when conjugation occur between
- | | |
|----------------------|----------------------|
| (1) $F^+ \times Hfr$ | (2) $Hfr \times F^-$ |
| (3) $F^+ \times F^-$ | (4) $F^+ \times F^+$ |

SPACE FOR ROUGH WORK

129. Viruses are not affected by antibiotics because
- (1) They have no metabolism of their own
 - (2) They have no cytoplasm and genetic materials
 - (3) They are obligate parasites
 - (4) They are nucleoprotein particles

130. Select the incorrect match :

Virus	Genetic material
(1) Cyanophage	DS-DNA
(2) TMV	SS-RNA
(3) $\phi \times 174$ phase	DS-DNA
(4) Cauliflower mosaic virus	DS-DNA

131. Viroids are small infectious particles which are devoid of

- (a) Capsid (b) Enzyme (c) RNA (d) DNA
- (1) a, b and d
 - (2) b and d
 - (3) a, b and c
 - (4) a and d

132. Which of the following statement is correct for Eubacteria ?

- (1) Mesosomes help in septa formation during binary fission
- (2) Rod shaped bacteria are called *Bacillus*
- (3) Teichoic acid present in the cell wall of gram positive bacteria
- (4) All of these

133. How many bacteria would be produced from single cell in culture after 160 minute if the generation time is 20 minutes ?

- (1) 64
- (2) 128
- (3) 256
- (4) 32

134. Nucleoid of Bacteria possess :

- (1) SS DNA + SS RNA + Acidic proteins
- (2) DS DNA + SS RNA + Histone proteins
- (3) DS DNA + SS RNA + Polyamines
- (4) DS DNA + SS RNA + Acidic proteins

135. For viewing diakinesis which one of the following would be a suitable material

- (1) Onion root tip
- (2) Leaf of *Dichanthium*
- (3) Tail of Rat
- (4) Bud of Flower

136. If a analysis is performed by a scholar on a piece of earth's crust & a living tissue. He obtain a similar list with a difference between two. Difference is :

- (1) In earth crust , Nitrogen is very little but silicon is negligible in living matter.
- (2) Oxygen is most abundant element in living matter
- (3) Four element (C,H,O,N) making 99% of living matter
- (4) All of these

137. How to analysis chemical composition found in living matter :

- (1) Grind it in Trichloroacetic Acid
- (2) Grind it in Trichloro acetylene
- (3) Grind it in Trichlorophenol
- (4) Grind it in Triacetone

SPACE FOR ROUGH WORK

- 138.** All the carbon compound/inorganic compound that we get from earth's crust can be called :
 (1) Biomolecules (2) Minerals
 (3) Genetic material (4) Humus

139. If a leafy tissue is burnt, which one is oxidized to gaseous form & is removed :
 (1) CO_2 , Mg (2) CO_2 , Na
 (3) N_2 , Water vapour (4) CO_2 , K

140. There are many amino acids, based on the nature of :
 (1) Amino group (2) Carboxyl group
 (3) R - group (4) None of these

141. The chemical and physical properties of amino acids depend upon :
 (1) Amino group (2) Carboxyl group
 (3) R-group (4) All of these

142. Which one is odd to lipid and amino acid :
 (1) carboxyl group (2) R - group
 (3) amino group (4) All of these

143. Nucleotides consist of :
 (1) Only pentose sugar
 (2) Only nitrogenous base
 (3) Only phosphate group
 (4) All of these

144. Some tissues, especially the neural tissue, have lipids with more complex structure. Which is it :
 (1) Phospholipid
 (2) Palmitic acid
 (3) Cholesterol
 (4) Sphingomyelin

145. The most common lipid found in cell membranes is Phospholipid, which is :
 (1) Simple lipid (2) Conjugated lipid
 (3) derivative lipid (4) Triglycerides

146. How many types of nitrogenous bases are found in nucleic acids :
 (1) One (2) Five (3) Four (4) Three

147. Which of the following constitute a nucleotide :
 (1) DNA only
 (2) DNA and RNA only
 (3) DNA, RNA and ATP also
 (4) RNA and ATP only

148. % of the total cellular mass, the principal organic constituent is :
 (1) Water (2) Protein
 (3) Carbohydrate (4) Lipid

149. Of the following, how many statements are true :
 (1) Protein is a homopolymer & cellulose is a heteropolymer
 (2) Cellulose contains complex helices, hence it cannot hold I_2
 (3) Lipids have molecular weights more than one thousand daltons and are usually macromolecules.
 (4) Sugar-phosphate-sugar is the backbone of a protein chain

Code :
 (1) Two (2) Three
 (3) Four (4) None of these

SPACE FOR ROUGH WORK

- 150.** Which one is the most abundant carbon form in the whole of the biosphere :
(1) Protein
(2) Cellulose
(3) RUBISCO
(4) Collagen
 - 151.** Biologist describe the protein & DNA structure at four levels. This statements is :
(1) True for protein, but not true for DNA
(2) True for DNA but not true for protein
(3) True for both protein & DNA
(4) False for both protein & DNA
 - 152.** Tertiary structure is absolutely necessary for the biological activity of enzyme because it forms:
(1) Co-factor
(2) Prosthetic group
(3) Co-Enzyme
(4) Active site
 - 153.** At the time of breakdown of 40 peptide bonds. How many water moiety will be released :
(1) 40
(2) 20
(3) 39
(4) 0
 - 154.** In biological world, glycosidic bond is formed by dehydration when :
(1) Polysaccharide come in existence
(2) Nucleic acid come in existence
(3) Polypeptide come in existence
(4) Both (1) and (2)
 - 155.** Number of chiral carbon atoms in α -D-(+)-Glucose is :-
(1) Five (2) Six (3) Three (4) Four
 - 156.** In primary structure a protein is imagined as a line, the left end represent by :-
(1) The first amino acid and its carboxyl terminal
(2) The last amino acid and its carboxyl terminal
(3) The first amino acid and its amino terminal
(4) The last amino acid and its amino terminal
 - 157.** Read the following statements and find out the incorrect statement.
(1) Oil have lower melting point (e.g. gingely oil) and hence remain as oil in winters
(2) Adenosine, guanosine, cytosine, uridine and thymidine are nucleosides
(3) Adenylic acid, thymidylic acid, guanylic acid uridylic acid and cytidylic acid are nucleotides
(4) Nucleic acids like DNA and RNA consists of nucleotides only
 - 158.** Polymeric polysaccharide consisting of only one type of monosaccharide is :-
(1) Glycogen
(2) Cellulose
(3) Inulin
(4) All of the above
 - 159.** Which of the following organic compounds are present in acid insoluble fraction but are not strictly biomacromolecules ?
(1) Nucleic acids (2) Polysaccharides
(3) Proteins (4) Lipids

SPACE FOR ROUGH WORK

160. Correctly Match the column-I with column-II

Column-I

- (i) Drugs
- (ii) Polymeric substances
- (iii) Alkaloids
- (iv) Pigments

Column-II

- (a) Gums, cellulose
- (b) Morphine, codeine
- (c) Carotenoids, Anthocyanins
- (d) Vinblastin, curcumin

- (1) i-c, ii-a, iii-b, iv-d
- (2) i-d, ii-b, iii-a, iv-c
- (3) i-d, ii-a, iii-b, iv-c
- (4) i-a, ii-d, iii-b, iv-c

161. Correctly match the column-I with column-II

Column-I

(Component)

- (i) Proteins
- (ii) Carbohydrates
- (iii) Nucleic acids
- (iv) Lipids

Column-II

(% of the total cellular mass)

- (a) 5-7%
- (b) 2%
- (c) 3%
- (d) 10-15%
- (2) i-c, ii-d, iii-a, iv-b
- (4) i-d, ii-a, iii-c, iv-b

- (1) i-d, ii-c, iii-a, iv-b
- (3) i-b, ii-c, iii-a, iv-d

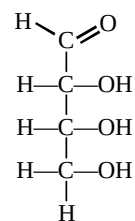
162. Which of the following biomolecules are not polymers ?

- (1) Lipids
- (3) Polysaccharide
- (2) Protein
- (4) Nucleic acid

163. Fructose is -

- (1) Laevorotatory
- (3) Disaccharide
- (2) Ketose
- (4) Both 1 & 2

164. How many chiral carbons are present in following structure



- (1) 1
- (2) 2
- (3) 3
- (4) 4

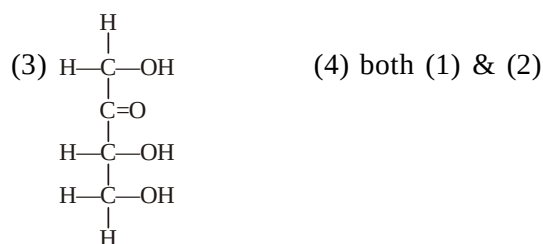
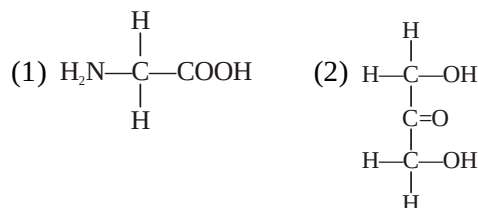
165. Ribose is -

- (1) Component of RNA
- (2) Pentasaccharide
- (3) 6 oxygen atoms present
- (4) Both 1 & 2

166. How many H-atoms are present in arabinose sugar ?

- (1) 5
- (3) 10
- (2) 4
- (4) 12

167. Which amino acid shows no rotation of PPL

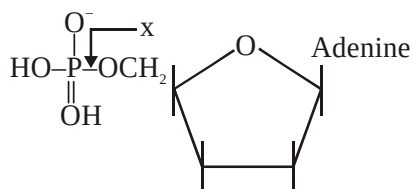


SPACE FOR ROUGH WORK

- 168.** Which of the following disaccharide has "1-4" glycosidic linkage ?
 (1) Cellulose
 (2) Starch
 (3) Maltose
 (4) All the above
- 169.** Which of the following will be, the correct words in place of A, B & C in given square ?
- | | |
|--------------------|---------------------|
| Deoxyribose | A |
| Sucrose | B |
| C | Ketose sugar |
- (1) A = $C_6H_{12}O_6$, B = Reducing sugar, C = Ribose
 (2) A = $C_5H_{10}O_4$, B = Nonreducing sugar, C = Ribulose
 (3) A = $C_5H_{10}O_5$, B = polysaccharide, C = Chitin
 (4) A = $C_3H_6O_3$, B = Monosaccharide, C = Glucose
- 170.** Read the following four statements (A-D).
 (1) Mannose is monosaccharide
 (2) Glucose is laevorotatory
 (3) Galactose has 12 H-atoms
 (4) Glucose & Fructose are isomers
 How many of the above statements are correct?
 (1) 1 (2) 2
 (3) 3 (4) 4
- 171.** What will be the number of hydrogen bonds in a 68 nm long DNA molecule if its adenine content is 20%?
 (1) 160 (2) 260
 (3) 240 (4) 480
- 172.** For a protein to have quaternary structure, it must have :-
 (1) Four amino acids
 (2) Two or more polypeptide subunits
 (3) Four polypeptide subunits
 (4) Atleast four disulphide bonds
- 173.** All protein amino acids are optically active except:-
 (1) Glycine
 (2) Serine
 (3) Threonine
 (4) Lysine
- 174.** DNA and RNA differ with respect to :-
 (1) only purine
 (2) only pyrimidine
 (3) Both purine and pyrimidine
 (4) Sugar and nitrogen base
- 175.** The bond formed as the result of dehydration synthesis reaction between two nucleotides is called as :-
 (1) Glycosidic bond
 (2) Peptide bond
 (3) Ester bond
 (4) Phosphodiester bond

SPACE FOR ROUGH WORK

176. Which type of bond is indicated by x in the diagram given below ?

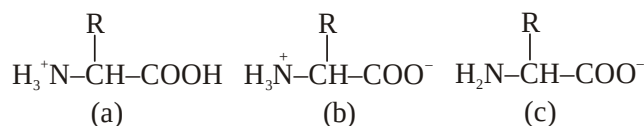


- (1) Phosphodiester bond
- (2) Ionic bond
- (3) Glycosidic bond
- (4) Ester bond

177. A segment of ds DNA has 120 adenine and 120 cytosine bases. The total number of nucleotides present in that segment is :-

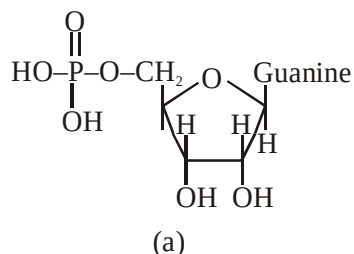
- (1) 120 (2) 480 (3) 60 (4) 290

178. Which of the following is a zwitter ion ?



- (1) a (2) a and b
- (3) a and c (4) b

179. What does a and b represents the structure provided ?



- (1) a = Adenylic acid b = Guanine
- (2) a = Uridine b = Uridylic acid
- (3) a = Thymidylic acid b = Adenine
- (4) a = Guanylic acid b = Adenine

180. ATP is :-

- (1) Nucleotide
- (2) Nucleoside
- (3) Nuclic acid
- (4) Nucleosome

SPACE FOR ROUGH WORK